MIL-P-45486B (PA)
13 May 1974
SUPERSEDING
MIL-P-45486A (MU)
5 October 1967

MILITARY SPECIFICATION

PELLETS, RDX
(FOR USE IN AMMUNITION)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers RDX pellets for use in ammunition (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-45	- Tape Gummed Paper Reinforced and Plain for Sealing and Securing.
PPP-F-320	- Fiberboard, Corrugated and Solid Sheet Stock Container Grade and cut Shapes.
PPP-B-636	- Box, Fiberboard.
TT-I-1795	- Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces).

MILITARY

MIL-B-2427	- Boxes, Ammunition, Packing, Wood, Nailed.
MIL-L-10547	- Liners, Case, Waterproof.
MIL-B-13239	- Barrier Material, Waterproof, All Temperature, Flexible.
MIL-A-13374	- Adhesive, Dextrin (Spiral Tube Winding for Ammunition Containers).
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For.

FSC: 1376

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

DRAWINGS

U.S. ARMY

8796522 - Marking Diagram and Sealing For Wood Packing Boxes.
9211789 - Ink, Marking, Opaque For Porous Surfaces (For Automatic or Semi-Automatic Machine Application).
9285526 - Pellet, Std. RDX.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 1-199

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402. Orders for the above publications should cite: "49 CFR 1-199 (latest revision)").

3. REQUIREMENTS

3.1 Materials. The constituent materials used in the manufacture of RDX pellets shall comply with all of the requirements specified on applicable drawings and referenced specifications.

3.2 Composition. The RDX pellets shall be homogeneous. The composition of the RDX pellets shall be as specified on applicable drawings when determined as specified in 4.5.

3.3 Cracks or laminations. The RDX pellets shall be free of all cracks or laminations when determined as specified in 4.4.2.1.

3.4 Detail requirements. Detail requirements pertaining to dimensions and pellet weight shall be as specified on the applicable drawings when determined as specified in 4.4.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. All pellets and packaging material shall be free of dirt, grease and other foreign matter.

3.6 Moisture. The moisture content shall not exceed 0.20 percent when determined as specified in 4.5.5 (see 4.4.3.1).

3.7 Density. The density of the RDX pellets shall conform to the requirements specified on applicable drawings when tested as specified in 4.5.6.

3.8 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of 100 RDX pellets obtained by sampling as described in 4.4.3. The sample shall be obtained from a production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials, shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.5 of this specification.

4.3.3 Rejection. See MIL-A-48078 (PA).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

4.4 Quality Conformance Inspection.

4.4.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each lot of RDX pellets shall be produced from one interfix lot of RDX powder produced in accordance with applicable drawings and specifications.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for the major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (See MIL-A-48078, para 4.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45486B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	RDX Pellets				(See applicable drawing)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Crack or lamination		0.40%	3.3	Visual
132	Weight		0.40%	3.4	Balance
133	Length		0.40%	3.4	Gage (CD)
134	Diameter		0.40%	3.4	Gage (CD)
Minor	None defined				

NOTES

5

SARPA QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45486B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Unit Packages				(See applicable drawing) NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Cushion missing		0.40%	-	Visual
132	Double cushion one end		0.40%	-	Visual
133	Contents loose		0.40%	-	Manual
Minor					
201	End cap not secure		0.65%	-	Visual/Manual
202	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45486B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Intermediate Package				(See applicable drawing)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Tubes not sufficiently separated		0.40%	-	Visual/Manual
132	Explosive content overweight		0.40%	-	Balance
133	Excessive moisture in packing		0.40%	-	Manual
Minor					
201	Carton closure insecure		0.65%	-	Visual/Manual
202	Marking misleading or unidentifiable		0.65%	-	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

1/

SARPA-QA FORM 2507 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

Quality Conformance Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-45486B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Wood Packing Case, Sealed				(See applicable drawing)
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major B					
131	Lumber undersize		0.40%	-	Scale
132	Board broken or split		0.40%	-	Visual
133	Contents exposed		0.40%	-	Visual
134	Strapping missing, broken or loose		0.40%	-	Visual/Manual
Minor					
201	Contents loose		0.65%	-	Manual
202	Strapping improperly assembled		0.65%	-	Visual
203	Marking misleading or unidentifiable		0.65%	-	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

4.4.3 Testing (Caution: see 6.5)

4.4.3.1 Sampling. A minimum of ten (10) pellets shall be removed from the beginning of the pelletizing operation and a minimum of 10 pellets shall be removed from the end of the pelletizing operation. The two 10 pellet samples shall be crushed and retained as two separate individual samples. These two samples shall be tested for the composition and moisture requirement in accordance with the tests depicted in 4.5. If either of these samples fail to comply with the requirement specified on applicable drawings or specifications, the lot of RDX pellets shall be rejected. If the total weight of either of the samples described above (beginning or end) is less than 100 grams, then additional pellets shall be selected from the pelletizing operation (beginning and end) to obtain a minimum weight of 100 grams of each sample.

4.4.3.2 Sampling for Density. When required by the applicable pellet drawing, a random sample of five (5) pellets shall be selected from the sample used in 4.4.2.1 and tested in accordance with 4.5.6. If any pellet fails to meet the applicable requirement, the lot shall be rejected.

4.4.3.3 Testing of RDX Pellets. The samples obtained in 4.4.3.1 and 4.4.3.2 shall be subjected to the tests listed in Table I.

Table I

Inspection/Requirement	Defect Classification
Determination of calcium resinate or stearic acid content (see 4.5.1)	Major B
Determination of graphite content of binder-lubricant (see 4.5.2)	Major B
Determination of total binder-lubricant content (see 4.5.3)	Major B
Determination of RDX content (see 4.5.4)	Major B
Moisture (see 4.5.5)	Major B
Density (see 4.5.6)	Major B

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

4.4.4 Inspection Equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose (see 6.2).

4.5 Test Methods and Procedures (see 6.3)

4.5.1 Determination of calcium resinate or stearic acid content. Transfer an accurately weighed portion of approximately 5 grams (g) of the sample to a tared medium porosity, sintered glass crucible. Extract the sample with three separate 20 milliliter (ml) portions of chloroform which has been previously saturated with RDX, maintaining each portion in contact with the residue for approximately five minutes before applying suction. Stir occasionally to break up any lumps which may persist and wash down the sides of the crucible with each 20 ml. portion of chloroform to insure complete removal of the calcium resinate or stearic acid. Aspirate the residue until the odor of chloroform is no longer detectable, place the crucible and contents in a drying oven at 100 degrees Centigrade (°C) to 105°C. for one half hour, cool in a desiccator and weigh. Repeat the above extraction, drying and weighing procedure with individual 20 ml. portions of chloroform until the loss in weight on subsequent extractions totals less than 0.003 g. Calculate the loss in weight of the crucible plus contents as percent calcium resinate or stearic acid, as applicable, on a moisture-free basis as follows:

$$\text{Percent calcium resinate or stearic acid} = \frac{100A}{W}$$

Where:

A = loss in weight of the crucible plus contents, in g

W = weight of sample on a moisture-free basis, in g

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

4.5.2 Determination of graphite content of binder-lubricant (when applicable). Extract the residue remaining after the calcium resinate or stearic acid has been removed with two 20 ml portions of dimethyl formamide, maintaining each portion in contact with the residue for approximately five minutes with intermittent stirring before applying suction. Wash the crucible and contents with a minimum of three 20 ml portions of acetone, maintaining each portion in contact with the residue for approximately one minute with continuous stirring, before applying suction and until the following qualitative test for RDX in the acetone washings is negative. Transfer the acetone filtrate to a 25 ml. beaker, evaporate to dryness in a steam bath and add several drops of a solution of chromotropic acid in normal (N) sulfuric acid. The formation of a reddish-brown color as compared to a pink color obtained by a blank prepared in an identical manner indicates the presence of RDX. Aspirate the residue until the odor of acetone is no longer detectable, place the crucible and contents in a drying oven at 100° to 105°C. for one half hour, cool in a desiccator and weigh. Calculate the weight of the residue as percentage of graphite on a moisture-free basis as follows:

$$\text{Percent graphite} = \frac{100A}{W}$$

Where:

A = increase in weight of tared crucible, in g
W = weight of sample on a moisture-free basis, in g

4.5.3 Determination of total binder-lubricant content. The percent total binder lubricant content shall be determined by adding the percent calcium resinate or stearic acid obtained to the percent graphite, when applicable.

4.5.4 Determination of RDX content. The percent RDX content shall be determined by subtracting the percent total binder-lubricant from 100 percent.

4.5.5 Moisture. Transfer approximately 5g of the sample obtained in 4.4.3 (weighed to the nearest .0001g) to a tared weighing dish with a cover. The uncovered dish, contents and cover shall be placed in an oven maintained at 100 ± 5°C for 1 hour. Cover the dish, remove from the oven, and cool the sample to room temperature in a desiccator. Determine the loss in weight by weighing the covered dish and contents. Calculate the percentage of moisture as follows:

$$\text{Percent moisture} = \frac{100A}{B}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

Where:

A = loss in weight, in g
B = weight of sample, in g

4.5.6 Density

4.5.6.1 Density. The density of each pellet shall be determined by using the following calculation:

$$\text{Density, g/cc} = \frac{\text{Weight, in g}}{(\text{Height, inches})(\text{Diameter, inches})^2\ (12.86)}$$

4.5.6.2 Density, alternate. The density of each pellet shall be determined in accordance with ASTM Method D 792-66, Method A-1.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A - Unit Pack. Unless otherwise specified, RDX pellets shall be tightly packaged in convolute or spiral wound chipboard tubes of the proper inside diameter to accomodate the pellets and not exceeding 15 inches in length. The chipboard tubes shall be four ply of .016 tube winding stock with dextrin adhesive, Specification MIL-A-13374, applied between layers. When spirally wound, all layers shall have butt joints with a maximum gap of 1/16 inch allowed. The inside of the tubes shall be closed with end caps of 0.25 inch cover stock. The tubes shall be sealed with one end cap glued to the body tube and the opposite end cap sealed with one layer of adhesive tape, around the joint of the cap and the body tube. The tape shall comply with requirements of Type III, Grade B of PPP-T-45. The pellets shall be cushioned at each end of the tubes with 3/16 inch thick pressed felt cut to the proper inside diameter of the tubes.

5.1.2 Level C - Unit Pack. RDX pellets shall be unit packaged as specified in 5.1.1.

5.1.3 Level A - Intermediate Pack. Tubes containing the RDX pellets shall be packed horizontally in a fiberboard carton complying with requirements of PPP-B-636, Style RSC, Grade 175, B Flute, Class Domestic cartons. The tubes shall be separated from the carton and between layers by a minimum of one inch of clean dry and tightly packed sawdust. Not more than 10 pounds, net weight, of explosive shall be packed in one carton. The carton shall be closed with two inch wide tape conforming to Type III, Grade B of PPP-T-45. Each carton shall be labeled to describe the contents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

5.1.4 Level C - Intermediate Pack. Tubes containing RDX pellets shall be packed as specified in 5.1.3.

5.2 Packing

5.2.1 Level A. One (1) intermediate package shall be packed in a Grade A, Type I, Class 2, wood box conforming to MIL-B-2427. Case liners fabricated in accordance with MIL-L-10547, Type I, Grade A using Type CW-2 Grade A material conforming to MIL-B-13239 shall be used in each box. The length and width of the liner shall be approximately 1/4 inch greater than the inside length and width of the box. Closure of liner shall be in accordance with instructions contained in the Appendix of the liner specification. After sealing the case liner, there shall be sufficient looseness between the contents and the liner to insure no pressure being applied to liner or seams. Both ends and the top of the box shall be lined with W5S or W6S fiberboard conforming to PPP-F-320. The box shall be closed and strapped in accordance with the Appendix of the box specification.

5.2.2 Level C. RDX pellets shall be packed as specified in 5.2.1 except box shall be Class 1, Grade C, and fiberboard filler shall be class domestic, Grade 175.

5.3 Marking. Each unit and intermediate package shall be printed, stamped or labeled with the following information:

(QTY) PELLET, RDX (1)

LOT (MFR SYMBOL - NO.)

PACKED (INSERT MO/YR NUMERICALLY)

HIGH EXPLOSIVE - DANGEROUS

(1) Insert No. of Pellets in unit package or intermediate package as applicable.

Letters shall be upper case and all marking minimum 1/4 inch in size. Labels shall be gummed, white paper. Ink shall be black conforming to TT-I-1795 Type II or III, or drawing C9211789. Each exterior container shall be marked in accordance with 8796522. DOT marking shall be HIGH EXPLOSIVES DANGEROUS and shall appear on the top and one side of the box in letters not less than 7/16 inch high. The words THIS SIDE UP shall appear on the top of the exterior container. Letter size shall be minimum 7/16 inch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. See MIL-A-48078(PA).

6.2 Inspection Equipment Designs. See MIL-A-48078(PA).

6.3 Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, ATTN: SARPA-QA-A-P, Picatinny Arsenal, Dover, N.J., 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.4 Intended Use. RDX pellets furnished under this specification are intended for use in detonators, leads, boosters, or any item requiring RDX pellets pressed from the compositions contained on Dwg 9285526.

6.5 Caution: Crushing of the RDX pellets should be carried out remotely behind a barricade.

Custodian:
ARMY-PA

Preparing Activity:
ARMY-PA

Project Number: 1376-A024

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:08Z
Check the source to verify that this is the current version before use.